Respiration

20 questions: 10 multiple choice, 10 structured.

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1 · Part 1 Core Respiration [1]

Which list contains only processes that use energy released by respiration?

A diffusion, growth, muscle contraction

B active transport, cell division, protein synthesis

C growth, osmosis, passage of nerve impulses

D active transport, diffusion, maintaining a constant body temperature

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2 · Part 1 Core Aerobic respiration [1]

Which statement describes aerobic respiration?

A chemical reactions in cells that make energy from oxygen

B the exchange of oxygen and carbon dioxide across a gas exchange surface

C the movement of air into and out of the lungs

D chemical reactions in cells that use oxygen to break down nutrient molecules to release energy

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3 · Part 1 Core Aerobic respiration [1]

Some germinating bean seeds are placed in a sealed flask. The flask is kept in the dark at 25 °C. The seeds respire aerobically.

After 24 hours, the air in the flask is analysed.

Which row shows how the concentrations of oxygen and carbon dioxide in the air have changed?

	oxygen	**carbon dioxide**
A	decreased	increased
B	increased	decreased
C	decreased	decreased
D	increased	increased

4 · Part 1 Supplement Aerobic respiration [1]

Which is the balanced chemical equation for aerobic respiration?

A $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

B $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 12H_2O$

C $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

D $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$

5 · Part 1 Core Anaerobic respiration [1]

Which statement describes anaerobic respiration?

A chemical reactions in cells that break down nutrient molecules to release energy without using oxygen

B chemical reactions in cells that use oxygen to break down nutrient molecules but release less energy

C the breakdown of nutrient molecules that happens only in organisms living where there is no oxygen

D the process in the lungs that releases energy when breathing stops

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6 · Part 1 Core Anaerobic respiration [1]

Which row describes anaerobic respiration in human muscles, compared with aerobic respiration?

	oxygen used	energy released per glucose molecule	product or products
A	no	more	lactic acid
B	no	much less	carbon dioxide and water
C	yes	much less	lactic acid
D	no	much less	lactic acid

7 · Part 1 Core Anaerobic respiration [1]

Yeast cells are placed in a glucose solution in a sealed container with no oxygen.

Which substances are produced by the respiration of the yeast?

A lactic acid only

B alcohol and carbon dioxide

C carbon dioxide and water

D lactic acid and carbon dioxide

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8 · Part 1 Core Anaerobic respiration [1]

During the sprint finish of a race, the leg muscles of a cyclist cannot get enough oxygen. Some of the muscle cells start to respire anaerobically.

Which word equation shows this anaerobic respiration?

A glucose → alcohol + carbon dioxide

B glucose + oxygen → carbon dioxide + water

C glucose → lactic acid

D glucose → lactic acid + carbon dioxide

9 · Part 1 Supplement Anaerobic respiration [1]

Some yeast cells respire anaerobically. They produce 40 molecules of ethanol, C_2H_5OH.

How many molecules of glucose were broken down, and how many molecules of carbon dioxide were produced?

	molecules of glucose broken down	molecules of carbon dioxide produced
A	20	40
B	40	40
C	20	120
D	80	80

10 · Part 1 Supplement Anaerobic respiration [1]

After a race, a runner continues to breathe deeply and quickly for several minutes. The runner has an oxygen debt.

Which statement explains the cause of the oxygen debt?

A Carbon dioxide has built up in the lungs and must be breathed out.

B Lactic acid has built up in the muscles and blood, and oxygen is needed for its aerobic respiration.

C All the glucose in the muscles has been used up and must be replaced.

D Alcohol has built up in the muscles and must be removed by the kidneys.

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11 · Part 1 Core Respiration [6]

A student investigates the effect of temperature on respiration in yeast.

She mixes 10 cm^3 of yeast suspension with 10 cm^3 of glucose solution in a test tube. The test tube is closed with a bung. A delivery tube leads from the bung into a beaker of water. She places the test tube in a water bath and leaves it for 5 minutes to reach the temperature of the water bath. She then counts the number of bubbles that come out of the delivery tube in 5 minutes.

She repeats the experiment at different temperatures, using fresh yeast and glucose solution each time.

The table shows her results.

temperature / °C	number of bubbles in 5 minutes
10	8
20	21
30	46
40	72
50	25
60	0

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(a) Name the gas in the bubbles. [1]

(b) Describe the effect of temperature on the number of bubbles produced. Use data from the table in your answer. [3]

(c) Calculate the mean number of bubbles produced per minute at 30 °C. [1]

(d) Suggest why no bubbles were produced at 60 °C. [1]

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12 · Part 1 Core Respiration [5]

Living organisms use the energy released by respiration for many processes.

(a) State two uses of energy in the human body, other than muscle contraction. [2]

(b) Root hair cells take in mineral ions from the soil, even when the concentration of the ions in the soil is lower than the concentration inside the cells.

State the name of this process. [1]

(c) Mice keep their body temperature constant. Scientists found that a mouse kept at 5 °C respired faster than the same mouse kept at 25 °C.

Suggest why the mouse respired faster at 5 °C. [2]

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13 · Part 1 Core Aerobic respiration [5]

Two vacuum flasks, P and Q, are set up in the dark at 20 °C.

- Flask P contains 50 germinating pea seeds.
- Flask Q contains 50 pea seeds that have been boiled and then cooled.

Air is pumped slowly through each flask and then bubbled through a test tube of limewater. Before it enters each flask, the air passes through a chemical that removes all the carbon dioxide from it. Each flask also contains a thermometer.

After 30 minutes, the limewater connected to flask P has turned milky. The limewater connected to flask Q is still clear.

(a) State the word equation for aerobic respiration. [2]

(b) Explain why the limewater connected to flask P turned milky but the limewater connected to flask Q stayed clear. [2]

(c) After two days, the temperature in flask P has risen but the temperature in flask Q has not changed.

Suggest why the temperature in flask P rose. [1]

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14 · Part 1 Supplement Aerobic respiration [5]

Liver cells respire aerobically.

(a) Write the balanced chemical equation for aerobic respiration. [2]

(b) In one experiment, a sample of liver cells uses 1.2×10^8 molecules of oxygen in aerobic respiration.

Calculate:

(i) the number of glucose molecules broken down [1]

(ii) the number of water molecules produced. [1]

(c) State the ratio of the number of molecules of carbon dioxide produced to the number of molecules of oxygen used in aerobic respiration. [1]

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15 · Part 1 Core Anaerobic respiration [5]

Rice plants are often grown in flooded fields. The soil around the roots is waterlogged and contains very little oxygen. For part of the time, the root cells respire anaerobically.

(a) Describe anaerobic respiration. [2]

(b) State how the energy released per glucose molecule by anaerobic respiration compares with the energy released by aerobic respiration. [1]

(c) Root cells take in mineral ions by active transport, using energy from respiration.

Suggest why the root cells take in mineral ions more slowly when the soil is waterlogged. [2]

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16 · Part 1 Core Anaerobic respiration [6]

A baker mixes flour, water, sugar and yeast to make bread dough. The dough is left in a warm place. The yeast respires anaerobically. The baker measures the volume of the dough every 15 minutes.

time / min	volume of dough / cm^3
0	200
15	236
30	278
45	314
60	340

(a) State the word equation for anaerobic respiration in yeast. [2]

(b) Explain why the volume of the dough increased. [2]

(c) Calculate the percentage increase in the volume of the dough between 0 minutes and 60 minutes. [2]

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17 · Part 1 Core Anaerobic respiration [5]

During a 100 m sprint, some of the muscle cells in the legs of an athlete respire anaerobically.

(a) State the word equation for anaerobic respiration in muscles. [1]

(b) Suggest why some muscle cells respire anaerobically during a 100 m sprint. [2]

(c) During a long-distance race, the muscle cells of the athlete respire mainly aerobically. The breathing rate and heart rate of the athlete are higher than when the athlete is resting.

Suggest how the higher breathing rate and heart rate help the muscle cells to keep respiring aerobically. [2]

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18 · Part 1 Supplement Anaerobic respiration [7]

An athlete runs as fast as possible on a treadmill from 0 minutes to 10 minutes. The athlete then sits and rests from 10 minutes to 50 minutes. The concentration of lactic acid in the blood of the athlete is measured.

time / min	lactic acid concentration in blood / mmol per dm^3
0	1.0
5	6.5
10	11.0
20	7.4
30	4.4
40	2.0
50	1.1

(a) Explain why the lactic acid concentration in the blood increased between 0 minutes and 10 minutes. [2]

(b) At 10 minutes the athlete has an oxygen debt.

Outline how the oxygen debt is removed after the exercise. [3]

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(c) Calculate the mean rate of decrease in the lactic acid concentration in the blood between 10 minutes and 40 minutes. Give the unit. [2]

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19 · Part 1 Supplement Anaerobic respiration [6]

Ethanol for use as a biofuel can be made by yeast. Sugar from sugar cane is dissolved in water and yeast is added. The mixture is kept in a large closed tank with no oxygen.

(a) Write the balanced chemical equation for anaerobic respiration in yeast. [2]

(b) After several days, ethanol is produced more and more slowly, even though the temperature of the tank stays the same.

Suggest two reasons why. [2]

(c) Suggest why the tank is kept without oxygen. [2]

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20 · Part 1 Supplement Anaerobic respiration [6]

Yeast can respire aerobically and anaerobically. A scientist sets up two flasks, X and Y. Each flask contains the same mass of yeast in the same volume of glucose solution at 25 °C.

- Air is bubbled through flask X.
- Flask Y is sealed and contains no oxygen.

Assume that the yeast in flask X respires only aerobically and the yeast in flask Y respires only anaerobically.

(a) State the number of molecules of carbon dioxide produced when one molecule of glucose is respired by the yeast in:

(i) flask X [1]

(ii) flask Y. [1]

(b) The yeast in each flask breaks down the same number of glucose molecules.

Calculate how many times more carbon dioxide molecules are produced in flask X than in flask Y. [1]

(c) Name the product of respiration that is found in flask Y but is not produced by respiration in flask X. [1]

(d) The yeast in each flask releases the same amount of energy each hour.

Suggest why the glucose in flask Y is used up faster than the glucose in flask X. [2]

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